Quality Control Protocol – Nikon Spinning Disk Microscope

Comments:

Necessary equipment:

- Printed copy of laser power template for the microscope
  - Or, laptop to directly fill in values in Excel
- Thorlabs laser power meter PM100D
- Thorlabs power meter slide-shaped sensor S170C
- Chroma slides
- Fluorescent beads: 170nm green (505nm emission), 500nm multicolor (430+515+580+680 emission)
  - Check that beads are not older than one year
  - For protocol, see: J:\Methods and Protocols\Sample preparation\ FluorescentBeads_updated250324.docx
- For cleaning:
  - Zeiss lens cleaning solution (Isopropanol + n-Hexane)
    - For recipe, see: J:\Methods and Protocols\Cleaning and maintaining optics\ The_Clean_Microscope_Zeiss.pdf
  - Dust blower
  - Cotton-tipped wooden sticks
  - Distilled water

<u>Start-up:</u>

- Create a sub-folder with current date in

  J:\Equipment\Microscopes\Imaging Facility\Nikon_SpinningDisk_N135\QualityControl; sub-folder name: yymmdd

- Switch system on:
  - Flip all switches on power strip behind the cameras
  - Flip all switches on power strip on right side of desk by PC monitor
  - Wait for initialization light on camera to stop blinking
  - Start up the PC
- Let the instrument warm up 60 min before start of QC measurements
- Clean all objectives and oculars:
  - Use the dust blower to remove loose particles from front and back lenses
  - Use distilled water and a cotton swab to remove water-soluble dirt
  - Use Zeiss Brillenreiniger and a cotton swab to remove oily dirt by moving swab in an outwards spiral

    - Inspect lens; if it is not yet clean, use Zeiss cleaning solution and a cotton swab. Again, use an outward spiral motion, from the center to the edge of the lens.
- Check all objectives for signs of damage:
    - Use inverted ocular to inspect the front lens and back lens
    - Use Front Lens Imager to make an image of each objective. Label it, and place it in a folder that will be move to today's QC folder at the end.
    - If not possible to use Front Lens Imager, use phone camera to take images of the front lens of each objective
- Check integrity of sample holders: Are there bends or scratches?
- Wipe dust and any spills from microscope and surrounding area
- Check that there's enough (and refill if needed):
    - immersion medium
    - EtOH
    - glasses cleaner
    - lens wipes
    - Kim wipes
- Note temperature in the room if possible:
- Check drives on the PC: Are they almost full?
    - Delete data older than 4 months
    - Make a list of file owners with substantial amounts of data, to ask if it can be deleted

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<u>Laser Power:</u>

Sample: ThorLabs power meter slide S170C

Objective: 60xOil; do not use immersion oil

Disk Speed: Does not affect laser power measurement

Procedure:

- Mount Thorlabs Laser Power Meter sensor S170C
    - Magnetic clips are too small to be placed flat across the slide, but can be attached to one side of the slide holder
    - Do not focus the laser on the sensor, because focus does not influence power measurement, but may damage the sensor
- Switch off room light
- For each laser and percentage on the template sheet:
    - **Switch to the OC of that laser (i.e., "S-488"); otherwise, filters may be incorrect and lead to misrepresentation of laser power**
    - Set the power meter to the correct wavelength

    - On the power meter, press Δ twice to set the reference value
    - Adjust the laser's percentage in the software
    - Press the green triangle "live" button to start scanning
    - Record the power meter value in µW next to that wavelength/percentage on the sheet
- Store power meter slide

<u>Camera Dark Counts:</u>

Sample: None

Objective: 60x

Exposure time: 1s, 500ms, 100ms, 10ms

Averaging: None

Camera: Both

Procedure:

- Switch to 60x objective
- Open ND_Acquisition and load the saved settings in D:/New Volume/ImagingFacility → "darkcounts.xml"
- Adjust file name: "darkcounts"
- Shut off room light
- For each camera, collect an image at the following exposure times: 1s, 500ms, 100ms, 10ms. These are pre-set as optical configurations in the "darkcounts" XML.
    - Change camera by going to Acquire → Select Photometrics… and then selecting the camera to be used.
- Save as darkcounts_cam4009 or darkcounts_cam4018

<u>Field Illumination:</u>

Sample: Yellow-green Chroma slide

Objective: ALL for main camera; 20x air for secondary camera (CFP, GFP, YFP)

Disk: Regular for all objectives; additional SoRa disk + magnifier for 60x

Averaging: 4 frames

Procedure:

- For each objective, use ND-Acquisition to set a multi-channel image with the one channel per wavelength: 405, 488, 561, 633
    - Adjust laser power and exposure time to avoid saturation
    - Take image
- Save a TIF for each objective as follows, adjusting the objective magnification: "field_illumination_**60x**_405_488_561_633.tif"
- Save image in J:\Equipment\Microscopes\Imaging Facility\Nikon_SpinningDisk_N135\QualityControl\yymmdd\
- Close projects

<u>Fluorescent bead imaging for PSF:</u>

Sample: 170nm green fluorescent beads

Objective: 60x, 100x

Averaging: 8x

Camera: Main

Procedure:

- Mount a slide with 170nm green fluorescent beads
- For the 100x and 60x objectives:
    - Find FOV with beads separated and located mostly in the center.
        - Should have approximately 20 beads in FOV
    - Find out suggested z-step; halve the step size to double number of imaged planes. Suggested step sizes are:
        - 0.1µm for 100x
        - 0.15µm for 60x
    - Set a dark-to-dark z-stack
    - Check that the image is not saturated
    - Image the stack using S-488 setting
    - Name file: 60x_PSF_488
- Save each ND2 file in J:\Equipment\Microscopes\Imaging Facility\ Nikon_SpinningDisk_N135\QualityControl\yymmdd\

<u>Fluorescent bead imaging for multichannel PSF:</u>

Sample: 500nm multicolor fluorescent beads

Objective: 60x

Averaging: 8x

Procedure:

405, 488, 633

- Set a dark-to-dark z-stack
- Check that the image is not saturated in any channel
- Image the stack
- Name file: 60xOil_multicolor_beads

<u>Check X-Y calibration of stage:</u>

Sample: 500nm multicolor fluorescent beads

Objective: 60x

Speed: 400

Averaging: 4 lines

Procedure:

- Open ND-Acquisition
- Check the “XY” option to enable selection of multiple imaging FOVs
- Mark two beads as ROIs
- Tune settings to avoid saturation
- Check the “Time” tab to enable a time series
    - Interval: -
    - Duration: -
    - Frames: 10
- Capture the time series image
- Save a TIF for each time series, named “pos1.tif” and “pos2.tif.”
- Save ND2 file in J:\Equipment\Microscopes\Imaging Facility\Nikon_SpinningDisk_N135\QualityControl\yymmdd\stage_repositioning.nd2

Axial coalignment of cameras:

Do the dichroics cause a shift in Z between cameras?

There are two options for dual-cam imaging at this scope: CFP/YFP, and GFP/RFP. But the beads we have don't fluoresce in CFP/YFP, so we only test the one combination.

Sample: 500nm multicolor fluorescent beads

Objective: 60x

Averaging: 4x

Camera: Both, dual camera imaging

Procedure:

- Find a field of view where several beads are visible, but not crowded together
- Open ND_Acquisition and load the saved settings in D:/New Volume/ImagingFacility → "dualcam.xml"
- Make sure Acquisition → Select Photometrics is set to Dual Cam mode
- Set up an ND experiment using dual-camera imaging
    - DM 561 LP should be the image splitter
    - Image GFP and RFP
    - Set a dark-to-dark Z-stack
- Capture the image
- Save the ND2 file as "dualcam_axial.nd2"

When there are other objectives frequently used, but not currently mounted on the scope:

- Mount additional objectives that have not yet been tested, but are regularly used at this microscope
- For each objective, repeat the protocols for measuring the PSF with 170nm beads and for illumination homogeneity.

Shut-down:

- Record temperature in the room:
- Shut down the PC
- Wait until the PC is completely off before switching off both power strips: the one on the right side of the desk, and the one on the air table behind the camera

- Clean objectives
- Check stage and surrounding area for spills, dust, etc.

Analysis

💡 Do these steps in order, because some sections rely on previous sections (exports from ImageJ are needed for the Python code).

Excel

- Copy the laser power template into today's QC folder
- Fill it out with values from the printed template

ImageJ

- Open the '60x_multicolor_bead' file
- Check that only one bead is visible in the image; crop it if necessary
- Open the 'metroloJ' Plug-in
    - We are using version 1.3.1.1.
- Choose the "Co-alignment" module (4th from the left, top row).
- Change the file name to coalignment_analysis.
- Set "Actual image bit depth" to 16
- In "Microscope Acquisition Parameters":
    - Set Microscope type to CLSM
    - Specify NA and RI of objective and immersion media
    - List the peak emission wavelengths for each channel
        - 405 → 430
        - 488 → 515
        - 633 → 680
- The defaults for Bead detection Options are generally fine
- Check "Apply tolerances to the final report"
- Set "Reject coregistration ratio above" to 1.0.
- Click OK.
- In today's QC folder, a "Processed" folder will be created by metroloJ. Move the PDF out of this folder, into today's QC folder, and delete the "Processed" folder.

MATLAB

- Make sure that the folder J:\Equipment\Software\MATLAB and all subfolders are listed in "Home → Set path"
- In the "current folder" window, navigate to today's QC
- For each single-channel PSF LIF:

- Run the following line of code (values are: filename, showPlot, zoomIn, redThreshold, showMicroscopeEvalPlot, showZProjection, beadSize, extractMetaDataFromPath)

psfeval ('60xOil_PSF_488.nd2',1, 1, 1.7, 1, 1, 0, 1);

- Save the figure showing “theo, ortho, pca” and the three mean bead images as a PNG titled “60x_PSF_488.png” (changing wavelength at the end for each channel)

Python

- Open Python script “QC_functions.py” and run it so that the functions are accessible by the Nikon QC script.
- Open Python script “QC_plots_nikon.py”
  - **Modules**: laser power, detector gain linearity, detector dark noise, field homogeneity, X-Y stage repositioning
- Change line 21 to indicate path to current QC folder, for the correct microscope and date:

  # Input the path to QC folder for a particular date:
  qc_path = r"J:\\Equipment\\Microscopes\\Imaging Facility\\ Nikon_SpinningDisk_N135\\QualityControl\\250321”

- Run the script.
- Check the QC folder for the following files:
  - laser_power.png
  - detector_gain_linearity.png
  - detector_darknoise.png
  - field_illum….png
  - stage_repositioning.png
  - PNG for each objective for the PSFs

Finishing up

- Copy/paste the Quality Control Results Summary Template to today’s QC folder.
- Insert all PNGs into the correct place of the template.
- Make notes of any follow-ups that should be done at the top of the template.
- If applicable, make a folder in the “LogFiles” folder of the microscope for issues that need to be followed up on.
  - Fill out a “Template - Hardware issues” Word Doc for each issue.
  - Email the microscope company if needed.
- Record today’s date in this Excel sheet: J:\Equipment\Microscopes\Imaging Facility\System_Overview.xlsx to track how often quality control is being performed.

- Email the users who have too much data stored locally on the system and ask if they can delete the data as soon as possible, or if the facility can delete the data for them.